

ActInf Textbook Group ~ Cohort 7 ~ Session 2 (Chapter 1, Part 2) ~ 7/22/20

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all right welcome back cohort 7 we're in
our second onboarding SL chapter one
slash Etc and Andrew go for it however
you want to
begin yeah uh
great so I know that
today we're covering not just chapter
one but just a more General like
introduction to
to the group and active inference um
sort of meeting uh I will kind of quick
quickly give like a rundown of this
first chapter it's not very long there's
not too much to say but um that said I
thought it'd be useful just to kind of
kind of wrangle together all these
different themes here um that we're
going to be seeing in chapter one but
also going forward for the rest of the
textbook so in chapter one we're very
quickly introduced to this general idea
that a we have uh agents which could be
seen as living biological organisms or
other kinds of self-organizing systems
um but they conceptualized as agents and
so agents take in sensory observations
and elicit actions to either realize
their desired outcomes or goals and
Andor to make sense of their world by
gaining information about it this leads
to the development of adaptive behaviors
or strategies that agents use for
navigating their environment uh this
process plays out in a recursive action
perception cycle where observations and

form agents decision making regarding the next action they should take uh we also learn that this this whole process occurs at various levels of complexity and differing time scales ranging from the simple and rigid moment moment by moment strategy I of nutrient seeking bacteria up to the more complex and flexible strategies of humans engaging in planning to realize their goals in the future active inference occurs at different physical scales as well as well from microbiology to Broad cultural and development developmental learning in collectives or human societies uh one way to frame active inferences approach uh for example is through the neats versus scruffies discussion in this chapter uh while so-called scruffies lay emphasis on the idio sequences and specificity of various phenomena of interests for example different biological adaptations neuronal Dynamics or cognitive mechanisms uh each deserving of their own separate bottomup research from empirical data the so-called Meats aspire to a set of first principles for explaining and relating these diverse phenomena in a more top- down fashion while still engaging in empirical analysis for validation and so active inference also identifies with needs as it lays the normative framework for understanding the brain and mind and their biological and cognitive implications alongside model design and testing for empirical analysis um we're

given a kind of General schematic of the rest of the book uh it's separated into two parts s the first half will focus on Theory the second half on practice so I'm just going to speak about the rest of the first half um the first half basically presents active inference as a normal normative framework with its distinct Concepts and theories um we're shown a kind of conceptual road map at how to arrive at active inference uh in you know your respective Learning Journey or however you would like to view that um and there are kind of two roads to reach active inference there's a low road starting with bases theorem uh which basically supplies the mechanics of active inference uh in its relation to what would be called the basian brain hypothesis uh it's basically a way of viewing uh processes or computations in the brain is playing out Thea basian principles where there's prior information uh new observations come in there's some kind of likelihood that relates uh prior information with new information and then uh produces a kind of posterior uh result you know a prediction about what's going on in terms of uh inference right inferring what is going on based on what you thought before as well as what new information you've received and then on the other hand the high road in chapter three uh coming much later begins with the fre fre energy principle or the

necessity for agents to minimize free energy as a proxy or upper bound on surprisal uh in order for these agents to survive and adapt to their environment uh for the rest chapter 4 we'll just go a bit more into detail about basis theorem and we'll actually be presented with uh major kind of Exemplar active inference models uh those that are used for discrete time scenarios as well as continuous time scenarios chapter five uh the final chapter of this uh theory part of the book we'll then uh discuss the neural process theory of active inference uh which you know again begins with the free energy principle but then also rate how the specific quantities we find in active inference like predictions prediction errors Precision signals and so on which we'll we'll discuss much later how all these map to cognitive functions and their constituent physiological substrate and neural computations so uh through the course of the first part of this book these first five chapters we just repeatedly returned to various themes uh perception as changing one's mind action is changing the world learning as actually being very similar to perception but operating at a slower time scale with longer term impact on the inference process going forward thus learning uh planning is a means of selecting actions and sequences to realize not just current but future goals by uh kind of kind of uh predicting forward into the

future to decide what you should do
later not just in the here and now uh
some other things like uh how to define
exploratory versus exploitative
behaviors it's a common distinction made
in various uh Sciences
Behavior Uh and and
I especially wanted to note those
because an active inference claims to
kind of uh include both kinds of
behavior within the same optimization
function for energy
minimization um as opposed to treating
them as uh separate and needing to be
manually defined uh as we'll find in
many other kind of computational
approaches like in reinforcement Le uh
agents for
example um
and that is about everything that I kind
of put together here for describing the
first part of the book um you know feel
free to obviously read the rest of the
chapter it'll talk about all the uh
other chapters in the book as well but
just focusing on part one for today so
um I guess I'd like to open up the
conversation I know that there's some
introductions at the very beginning of
the meeting so I'm not sure if we want
to go back into those but uh usually how
we do it in the textbook is just anyone
who has uh something of interest uh
about this chapter or a question they
have um they would just like to throw it
out there we could uh take things from
there there there's things that um I
have to say but I'm look in my not I I I

make notes on my trell board so if anybody else has anything to say before I I get to the notes I've taken on this chat on um I guess they could feel free I'm just looking for my notes sure uh Jorge you have something yes basic question uh what is free energy itself like not the pr principle but free energy yeah so it's uh For Better or Worse is somewhat uh complex concept but um it's it's basically it's you can relate it to information Theory it's kind of like an information theoretic quantity if that makes any kind of sense that Mak sense yeah so it could it could be measured in NS for example um and kind of the the importance of it here is that it acts as a kind of proxy more specifically a kind of upper bound on another quantity that organisms are theorized to be trying to minimize called surprisal and the reason why we're not um try saying it's surprisal minimization as opposed to free energy minimization so surprisal as far as kind of the basing computation that's being played out here surprise is typically like intractable um it's just a a lot of um kind of factors that need to be marginalized along the way when when Computing surprisal there can be kind of this issue with integrals and so on and

so free energy in kind of mechanical terms is just yeah an upper bound on surprisal that is tractable so we can compute it then on the other hand um it still relates to something like something like a kind of biological realism that is you know uh organisms have limited uh resources in the Moment by moment inference processes that they undertake right so they're biological and metabolical um kind of constraints on on what they're able to do hopefully that was helpful

yeah let me add one more by by analogy here so from chemistry one free energy type that people might be familiar is the Gibbs free energy so this is the thermodynamic free energy and so it's like ball rolls downhill is the um Newtonian mechanical path of least action given gravity candles burn given the activation barrier is the thermodynamic equivalent so put putting aside this the activation energy question going downhill means energy is dissipated as heat but then there's some floor where there's no more liberable energy from that floor so free is not in terms like free like beer free like software free a synonym for it is like liberable or like releasable as heat so then 1900s are happening information and thermodynamics are coming closer in info thermodynamics with in a lot of different like kind of

really mechanistic ways like Land hours
limit how much heat is related to like
erasing one bit but also like
realizations that at even deeper levels
that there's these um way to think about
information in terms of also a path of
least action in an arbitrary State space
that's called basy mechanics so the ball
rolling downhill in this case is like
explaining the data better or better is
given a specific like form of families
equations such that there's kind of a
signal to noise and the signal is kind
of like the the floor and the noise is
more like a heat or an
entropy so it's used as an approximator
that bounds surprise minimizing surprise
is maximizing model evidence however
maximizing evidence you don't know how
high that mountain is going to go but
minimizing surprise is often more
tractable but then it itself is
intractable for another reason which
necessitates in the approximate basian
setting the usage of variational methods
because that gives an
analytical approach that's optimizable
incrementally whereas in alternative
approach to variational approximation is
Monte Carlo or sampling based methods
those are also Limited in large State
spaces but they don't require you to
compute a free energy to balance a prial
you just sample from a distribution and
then use that as your kind of empirical
data
David uh yeah I was wondering if we
could correlate it to to um

free energy in in terms of tradeoff so
like how much how much weight is
assigned to you know the predictive
nature of tradeoff itself and how much
will you gain or lose by being right or
wrong you know depending
on the the data itself and how it's
fit the accuracy minus complexity
decomposition of variational free energy
energy energy minus entropy is kind of
like the physics e flavor accuracy minus
complexity is the statistical flavor
which is that you want to reward model
fitting but penalize more parameters so
now your kind of your risk tolerance
like whether you have a preference for
smaller and you're willing to like your
point of diminishing returns is a
smaller model or whether your point of
diminishing returns is a larger model
there's like this parto Optimum
optimality manifold where if you're off
this manifold you just could simply be
doing better but then there's like a
manifold Frontier in terms of model fit
where it's like it's like a a
hyperparameter manifold where you're
doing as well as you kind of could on
that trade-off Frontier but off that
crease you're just leaving um
explanatory power on the table with the
parameters you have
or or from coming from the other side
well that's kind of the
statistical and then this is part of the
whole theme with things the the the most
likely thing is what happens like things
are doing what's

likeliest rather than what's most rewarding because the proposal of a secondary reward function is not even something that seemingly Advanced cognitive systems necessarily do but that's what gives Unity with the dust particle and The Rock and cognitive modeling but then the question is how do you explain higher order Behavioral Behavior Etc in terms of novelty Aesthetics in Surprise minimization

terms right so I mean I guess sorry my my approach to it you know in terms of like I come from more of like the quantum side of things so I don't want to inject too much of that into it but um if you know you have a coupling with like a bit flip right if you have a coin with two heads like if you're just going off the randomness of the coin flip if you're a blind Observer you don't know you know heads or tails what the trade-off is going to be I mean if it's given the outcome is always going to be heads how can you then minimize that into free

energy well if you're never getting an observation then you're never able to calculate a free energy because the free energy is C equation 2.5 for variational free energy it's your beliefs and an incoming data

points so if you don't have a data point you could do descriptive statistics on belief but you wouldn't calculate a free energy

quantity right so I guess the I guess

the primary question there is if a system is determining how you're going to answer something because of you know learned in

Behavior due to an

assumptive prere is it doesn't that kind of end up being deterministic and

becoming a trade-off in the long

run I'm sorry can I um make comment

regarding like my the way I think about

free energy and how it makes sense to me

like I like to think about Fields I'm

assuming everybody on everybody here

knows what a field is and I think about

the different combinations of ways that

perception and action can unravel with

perspective prior thinking about feels

from the physics point of view and

translating that to the

cognitive and Behavioral or

psychological point of

view um and you can get really more into

the woods with this in terms of like you

know mechanisms in the brain where

neurotransmitter interactions are

happening and neuromodulator

interactions are happening but um I I

I'll get into like certain points that I

made about comments I made about the

chapter in a moment but I just figured

that what makes sense to me about free

energy and just thinking about like

potential

energy in from from a a the point of

view of like how energy can be utilized

for it's utilized and think about the

degrees of freedom that's that's there

and thinking about modeling agency and

how organism

um engage their environment but also um

a number I've had a number of com

conversations regarding free energy

being generalized beyond the scope of

thinking about agents or organisms but

also in their objects and how they take

up an environment and how you can

estimate the space over time that's

taken up by those objects and then you

can encode the marov blanket and what

have you have a field you can envelop a

field uh of like the possible

interactions that can happen even from

inert objects to um systems that are

capable of agency that can actually

properly

infer and estimate over

time um uh their environment and their

circumstances leveraging things so I uh

to me that that it makes sense from that

point of view and not to not delving too

much into the details of the physics but

acknowledging okay there's the concept

of fields and I could take that in the

combinatorial options in which me as an

agent can perceive and act on my

surroundings and develop priors but

hopefully that that that kind of uh

is applicable to addressing this like

potential energy free energy so and what

have um mean to ramble too much but I I

I was like sitting on that thought for a

few minutes and like oh I want to say

something and

hopefully it helps make a more

sense yeah I mean it's I mean not to

heavily I'm really just kind of

simplifying and abstracting certain parts of what you said so apologies for that but I to kind of bring it back I think I mean if I understand David's question right um like isn't maybe part of the idea is like isn't supplying some kind of Prior that's already heavily conditioning the the process that ensues from there such that there's a kind of almost determinism based on your initial starting point because we're requiring that you have a starting point given this is like a Bayesian process where have prior information necessarily involved and I mean

it's from my understanding it's just kind of like a description of how the process plays out and I like the idea of a field whenever we just look at it as like this is a kind of space like how you describe like like objects or beings in their environment their you know uh affordances within that space or field or environment and so on and it's like well if we were to study that as you know scientists researchers however like we already have a lot of starting kind of prior

there and we're going to start trying to model it to predict what will happen next so I mean

it's not to sound so abstract or conceptual but it's like we're always kind of working with from our standpoint kind of you know

external uh position a bunch of Prior information and we just kind of start there um just a couple tangential points

that uh when setting up an active inference model um you know there there's of course like fitting models to empirical data but then there's also just during the testing phases I mean it's very common to just initialize with say very uniform uh priors you know B basically entirely uninformative priors and it's through the capacity for like learning or updating model parameters that the the agent can actually adapt to the situation and then it's it's it's its own parameters change such that you know it's kind of starting with new priors in in the next round so to speak that may be better hopefully through Min free energy minimization better match the environments around it um if any of that tracks yes and just like dynamical systems or linear aggression you can pull back to like univariate linear regression or single sinusoidal oscillator dynamical system those are kind of like the doorway example but then you might run a statistical script and it does a linear aggression on thousands of data points or it h it has more Dimensions than a single dimensional form so in the textbook much as it is for for for again the kinds of um topics that are addressed um in the appendices linear algebra dynamical systems probability information Theory those are also like their

principles are shown in the textbook but then when it really comes down to anything like a system of interest it's the shape and the specific variables you use so these are like some of the moves and some of the generic cues and then like it's all about making that jump or path from okay listening to music in chapter 7 to here's this music data I have yeah that makes sense I I guess I'm thinking in terms of Weights as well like if we're waiting that data um like how can you avoid overfitting on something that's assumptive like for example like some of the things covered in chapter one like do go into I guess some of some of the things I see as um like in terms of like not just tradeoffs but in terms of how we approach as you were saying like just with sensory response right like our priors are also based on our limitations in terms of how we can measure how we can even understand to measure where you know there might be extra sensory kind of abilities that we have from a cognitive standpoint that we're not aware of in terms of even just in the way our brain interacts with itself with neuron interaction right so um I guess it goes back kind of to the pheromone response as an example of something that we can't detect but should be able to be weighted in the data yeah I mean just as a a brief like like one of the key

a key Concept in in active inference
whenever it comes to modeling is the
notion of like precision as a kind of
inverse variance uh term and that that
can kind of be viewed
as a weight or waiting system there are
a variety of different kinds of
precisions that are applied to different
variables as far as the neural process
Theory goes they're kind of related to
synaptic gain and and neural trans
mission um but uh yeah the general
sense is that there's a kind of waiting
process going on in addition to that
again active inference uh agents who
engage in learning like many things can
be learned including like how to adjust
those
precisions uh in order to say weigh
upweight or down weight the impact of
new incoming
information um you know there there
might be an agent who uh has a
likelihood model that that Maps
observations to its own beliefs about
what's going on is that kidden states
and so it it might it might have a lot
of really
imprecise mappings like relations such
that even though it receives new a new
observation it kind of downweights the
impact of that new observation because
it
it can't it can't find a good way of
mapping it to a to a belief State such
that it can confidently update that
belief State giving them
observation so so there are these kind

of mechanics

yeah great point with gain that can be

thought of like people with audio

editing experience probably have a very

embodied sense of this but there's a way

to tune the Precision such that even a

variable signal gets collapsed to

basically the extreme case is the dero

Delta function which is just like a line

it's a it's a special distribution that

is zero everywhere and then it's

infinite in one spot so there's a

Precision setting that'll take a range

of values of thermometer readings the

height of the children in the classroom

and do something like collapse it down

to its central tendency and then also

there's gain settings that can take even

closely related data points and flatten

the

distribution the hyper priors and some

of the parameters that that come into

play in empirical basian analysis and

then like unless otherwise specified

it's like the temperature is

one but then sometimes you do write it

out more fully and then H learn or

manipulate the hyperparameter related to

temperature and then about action that's

the shaky hands so it's kind

depending on where it is but but that

does basically like sharpens or

blurs okay yeah I mean that's how I was

thinking of it in terms of like gradient

descent um because those are some of the

formulas that I've been you know working

on where I also have run into a few

things that I need to check where it is

zero like you were mentioning but it also expands infinitely at that point and contracts infinitely at that point so that's kind of my prior that I'm trying to work with in terms of like if the math is correct what inference does that have if that collapse point will always be zero and one right you have a binomial distribution at that point where it's still a choice but if it's predetermined right you end up and I want to get too far into the woods on that but it's just something I wanted to kind of highlight from what you mentioned I want to um bring up so there is one statement on page 10 that I think I have some confidence that's that that that uh the rest of the book kind of mix up for for for the generalization I found which uh is the this the the the the bullet I think it's U it's the second it's the last bullet on the on page 10 and and it's it's a bol that says under active inference both perception and learning are active processes and I quoted the part the brain is essentially a predicted machine which constantly predicts incoming stimuli rather than passively waiting for them and the comment I made about this was that I thought it was a it was it was kind of a naive assumption to assume that there's never a situation that the brain is passively engaging with certain stimuli instead of constantly predicting and this can help differentiate between

noise and relevant stimulus to be interpreted as information so I think that the rest of the book gets into better detail about this but um I made a note of that as like kind of um a naive assumption made on page that I I found on page 10 that um well is the brain predicting everything coming in around it's predicting relevant information but I I guess I'm assuming that the book gets real uh it seems like I I Peak that chapter to little but it seems like it's it's get it does it better at spe better job at specifying and I think that overall chapter one does a good job providing a nice General overview even though there's there's some assumptions made that I think are clarified better but um I also sorry about that yeah point I made about what I found on regarding predictive processing happening in the brain yeah two two quick things on this the first is as so so so many times there's slipping between realism and instrumentalism if they said first we choose to model the brain as a predictive machine they couldn't be wrong because they're talking about their position to modeling but then it slips into what the brain is so that's that's one piece and the second thing is it hinges on what is meant by passively waiting that could be hanging out on the couch but hanging out on the couch even if your eyes weren't

cating still you would be actively
generating the periphery the sharpness
and so on so in that sense there is this
real time like now casting a vision even
if it's not cating and moving its head
and all of that but then is that passive
or

active yeah I mean how does that speak
to a dream State as well where if you
have a hyperreal dream state you know
where there's you can affect your
surroundings in the dream state not
realizing you're in the dream state
versus awaking state in the same kind of
notion of

sensory yeah well that I mean that's an
interesting question I the way I would
take them is that it still be similar to
a waking State and that you're still
a bunch of sensory information is still
kind of being delivered to you
presumably there's something going on
that is kind of actively leading you to
make the decisions or inferences that
you're making during the the dream um
but I've never actually I've never
really followed that full thought
through on thinking about how this works
in a dreaming state but that's just my
initial guess at an answer to that
question um I think for this particular
quote yeah same point that Daniel made
earlier the flipping back and forth
using instrumentalist language sometimes
other kinds of language other times but
I I just wanted to say I think it still
relates back to kind of that you always
have priors right there's that there's

that kind of like you have prior there
and then there's computation going on if
you didn't have priors then maybe it'd
be easier to say like like you could
passively take something in but the idea
is
that you're you're coming to a situation
figuratively with information that's
going to get computed so it's like
you're involved in the process as it
works and to make this like really
clear and be clear about what the
alternative hypothesis to a Bayesian
approach would be frequentism would say
okay we're getting this wave of photons
hitting our retina and we're going to do
the maximum likelihood
reconstruction so then we'll use
interpolation gaussian blur
convolutional neural net etc etc etc all
these techniques that are these inbound
processing techniques to reconstruct a
get a snapshot of the maximum likelihood
image so that is implicitly saying all
pixels have a uniform prior like I don't
know anything and I'm just going to take
one snapshot of pixels and then
reconstruct the maximum likelihood outcome but as
soon as you start to leverage oh well
but I think it should be like the prior
one prior
frame that's literally Bayesian analysis
then that's the Kalman filter where on
one extreme it is the special case where
you're frame by frame doing a total
reestimation The Other Extreme case is
you're just making a longterm running
average blurring over all the times so

then you have this Bayesian hyperparameter search question how responsive should my Colman filter be and then you have another model in this portfolio you know how fast should I be learning how fast to change that all of those Bayesian models can be evaluated with Bayesian model selection using accuracy minus complexity Bayesian information criteria so it's like oh okay well I'm getting I'm still getting bang for the buck to three but four always you'll fit better and better but then you're past the point of diminishing returns so that's like kind of the meta with Bayesian which comes back in chapter nine as well is selecting amongst Bayesian models even with totally disparate architectures is a Bayesian model selection question whereas in frequentist parametric statistical methods you can really only compare model likelihoods using strict AI nested the hierarchical likelihood ratio test Anova but you can't test non strictly nested models so for all the limitations and peculiarities of Bayesian analysis it's super useful to compare it to frequentist and to non-parametric statistical approaches like bootstrap jack knife resampling but those are three of the major lineages in statistics overall and those are some of the major toolkit areas as well and the SPM textbook earlier work

by friston go goes into this and kind of
compares the methods a
lot so is the stacking kind of like
logarithmic in a sense or is there any
point where it becomes exponential and
then just falls back through the stack
to the base
what
stock well so you were saying like if
you're using basian to calculate basian
right so you're still you're kind of
using an exponential formula in a way
right but it must be within a certain
range so once you hit a maximum does it
reduce back into the root you know it's
still basian inference like you were
saying right as a four function
or would it reduce all the other layers
that you've gone through to iterate up
into a certain point right back into the
initial if there's no priors in the set
okay is an interesting question so this
is like asking what should your prior be
over structurally variant models like I
I can do the one layer temperature model
just try to do a fit I could do a two
layer temperature model three for but
then am I a
frequentist which is to say naive
over structural
parameters okay so is it like is the
whole basian game nested within a
frequentist navity about model structure
and you could do that or you could
explicitly have a prior on structure
learning so then if you were saying I'm
making structural models of this
condition in the genome and I I think

it's it's not possible for there to be fewer than five or more than 50,000 low side and then here's my distribution of of structural parameters within 5 to 50,000 it's uniform within 5 to 50,000 or it's gaussian around this number then your structure structurally differing models would be weighted attentionally by your prior Distribution on that okay whereas if you have a uniform distribution across structurally different models that's like a fair test across the all okay that makes sense thanks yeah I had a one more comment on the chapter on page 10 the second bullet where it starts with various cers of active inference have possible biological analogues in the brain it makes they claim that Precision of policies corresponds to firing rate activity so I think again that that's a pretty big assumption and there's a lot going on with the regulation of neurotransmitters and other main neuromodulators like nor epinephrine and serotonin so the regulation of arousal in the brain and um the coupling the moments of coupling between serotonin and dopamine for for different reward uh learning circuits that are that are chained together as well so and I think this also relates to uh the capacity to become aware of and pass on stimuli as Sensations and acting on those Sensations that impacts perception and action and priors so I I figure I I

throw that out there
too but I I'm not sure totally
if the if other chapters in the book
press on the importance and role of
other neurotransmitters and neurom
modulators from from the top the of the
stack in terms of the of how cognition
is uh uh is
regulated within the
agent yeah here's table 51 acetylcholine
nor adrenaline and it goes it here's the
citations with some other
neurotransmitters and and um agreed it's
like it's like but there's there's so
many subtypes receptors Downstream
signaling it's like how could one single
parameter and it's like it's a course
graening it's a map it's not a territory
it's not that it is a parameter in the
brain and then models that make unique
predictions and explanations and have
explanatory value those are leading
models and then when they when they are
fit with one data set and then they have
predicted value in a different Paradigm
again that's evidence for that it's
capturing an informational
aspect so yeah I mean thinking about
that in terms of like descending versus
ascending and deep versus superficial I
mean that's kind of the coin flip again
right where you have like at what
point in that chain of the
structure can you Det determine whether
or not the Deep has flipped to a
superficial kind of
state if we're thinking of temperature
right it's going to be a constant right

or are we talking about interval in this case temperature could be constant or just like from reinforcement learning and other basing approaches a common method is to have a high temperature beginning and then turn down the temperature and then people say well let's do multi-chain let's have them do let's do five parallel chains with different temperatures and then have them swap positions like then there's all these higher order questions about how you best leverage your Computing resources but that's not changing the analytical formalisms those are more like computer science and engineering methods to Implement variational basian methods okay like that doesn't change the equation of 2.5 or 2.6 but like different approximators they may give different results but it's like that's like how many layers back do you pull back in your investigation right question which what what's the name of the book that you mentioned about by friston on the SPM I think it was earlier what's the so the last um printed version as far as I understand was 2007 it's purple book okay but then there the the SPM online documentation goes up to 2012 and there's still a totally live Community with people doing fmri

Imaging so I found this I found this
book to be extremely helpful background
information for several reasons first
off it's like a speed course on the
trifecta of classical parametric
statistics non-parametric statistics
basian statistics so it gets all those
in play all the time and it deals with
what we might call the a matrix the
mapping between observations and hidden
States in a setting where the
observations are of a well-known form
like EEG or fmri sensors and then the
hidden states are neural activity and
the blood oxygen flow dependent signal
the Bold fmri signal or neural activity
signals so it's like a defin system of
interest and sensor type clarifies the
setting it's not about um
multiscale um considerations but there's
some some ways towards that um so it
really hammers down the statistical and
the partially observable
part and then action selection is just
selecting a different index card in the
transition operator and changing how the
hidden State
unfolds and and that is kind of
approached towards the end of the book
in certain
analyses and it highlights the empirical
basis from your Workshop presentation
Andrew what what questions did you
receive or what did you feel like what
what what would welcome people
and um like what what did people ask and
what were they wondering when you showed
them what you

made yeah sure um yeah you know it's so again the the conference was uh it's it was like the big like International uh like computational social science conference and so many people there are like familiar with machine learning and and empirical data and so on maybe not know so much um kind of like say foundations of Concepts in physics um information Theory and so on um not in a way that they firmly needed uh and so I think a lot of it's just attempting to answer questions of like why use active inference to try and explain human behavior can we not approximate human behavior through you know iety of other methods um so it's it's a little different from the conversation we've been having right like there's a lot of questions of like scalability like oh how okay great one person one individual entity is minimizing their free energy but how does that allow us to make claims about what's going on at say more Collective you know level like like 30 agents in a network or something like that um you know so it's kind of just um it's more about kind of describing the mentality here of like okay we're we're creating something like an agent based model that has an agent and its environment and then from there it's just you as opposed to using like net logo which is kind of traditional way for scientist social scientists to do this where you have much more rules-based logic like the

agent does X if it observes y the Z if
it observes a um you know and said it's
kind of like we have a models in a model
so we have an agent based model where
each agent is its own generative model
um so yeah it's it's just more like kind
of conceptual in that sense how to
relate it to agent based modeling human
behavior
especially
um let see what
other you know it's per usual there
there was a quick question about like
okay so why the is this a theory of
Consciousness Etc you know that they
kind of open up into other
discussions um let's see yeah I think
sorry about those massive cells there
but yeah those those outputs are like
you know this is
Agents engaging in something like
Collective problem solving so exploring
an
enormous landscape where they can either uh
explore which is where they change a bit
in their own solution or they can
exploit meaning take their neighbors one
of their Network neighbors Solutions and
so the these are just like the output
results for running that over T time
steps right like how
how what what's the best solution that
they managed to collectively come up
with did all agents arrive at that
solution is it separate you could try
and use this to study something like the
the diversity of ideas in a more General
way too of like right because you know

there are fewer and fewer unique
Solutions if all agents are just jump in
to stealing the best one um so that yeah
Falls over time and then what I was
attempting to demonstrate that because
this is recreating a simulation of more
famous kind of Paradigm uh from a decade
or so ago where they try and describe
like team performance between um you
know within groups like a group of
researchers or a group of you know
employees or something all working
collectively on a problem maybe it's a
group of students right working on a
group project um but what they don't you
know it because it's all done in this
kind of like more traditional
rules-based way we're not able to look
at like okay well what about agents who
learn you know and actually adapt right
and I'm trying to demonstrate that
whenever we include these agents is
active inference agents who can learn uh
actually like the impact of engaging in
a project where you're able to just
steal your neighbor's answer without
doing much of the work yourself like if
that's kind of what you learn or adapt
to then uh then over time like that's
kind of where you'll start off your next
group project potentially like your
beliefs about what the optimal solution
is Will point you towards doing the
easier thing so so it's just kind of
trying to add a lot of more Nuance to
those that traditional simulation it
recreates the previous simulations
Dynamics it recreates the Dynamics found

in empirical literature of of group uh
Dynamics and performance and yet this by
creating agents that have these kind of
internal cognitive active inference
parameters you can you you know the idea
is to try and have proxies for like okay
what how does this impact back on the
agents themselves and the longevity of
doing practices uh practices like these
so so those like those are just some
animation plots that show like the
distribution of beliefs of uh if it's
basically mapping the likelihood of if I
see myself like exploring coming up with
my own answer you know what how does
that map to observations that are oh I'm
getting a better answer so it's kind of
like a personal
efficacy um belief it's like oh how much
do I associate exploring with a better
result a better solution in the end
which then would condition your
decisions uh over your actions like your
posteriors over policies to continue
exploring right so over time they all
start off you know kind of low that's
how I I set them up as priors but many
of them just get lower and lower like
they you know you just you have so many
neighbors that you can kind of take the
answer from so I I could go on about
this all day but it's just you know the
the the slides get into some of the more
kind of
textbook formulas equations Etc but the
the the script here was really to try
and demonstrate like what happens at a
broader scale and how could you actually

use this

um you know to approach problems that we're already trying to get at like not throw a bunch of new stuff at people all at once um about here's a whole new set of problems you can now try and approach thanks to these methods instead it's like here's the value added of using these methods even whenever you try to approach um you know existing paradigms or problems so that's

my I ask you a question

um I haven't had a chance obviously to look at the data but um you had mentioned you know like uh a group of like 30 people for example in a node if you consider that you know one company right let's say it's a department a company you're still connected to another node of say another if your n value remains 30 per

Department um if you had you know one individual in each of those nodes that wasn't conforming or wasn't you know kind of going with the same flow was there any correlation did you see it anywhere else in the topology that might have affected chains that aren't directly connected but would be through the network itself that's interesting

um so I didn't get quite that far right because this was basically just kind of put together as like a toy situation right just to demonstrate but that that's an interesting question I think you could attempt to get it something like that like so this is

basically just creating a single Network
like a a single
um you have two options like the way I
set up the function you can either make
a like a random graph with like P
connectivity that you can Define
yourself or you can set it up as like
subn networks um where where the the
separate subet it's like five groups of
four agents or however you you know want
to Define that but they're completely
separate from one another
but what you could do is you could
adjust that to where you know you do
have connections between the sub
networks um and then I wrote everything
you know just in a very programmatic
like make all the agents have the same
priors but you could set up an agent who
has you know very different priors from
the rest see how that plays out so I
guess the the simple answer is like I'm
not sure but the the the the tools are
kind of there if one wanted to maybe
approach and answer to that
question great thanks
sure cool maybe a model stream maybe
with somebody else who's um done it but
you should post it in Discord and let's
see if anyone plays with it and then
let's do it because I think that's a
great framework
for playing around and linking it to the
sections of the
book sounds good yeah I'm I'm I'm
someone who comes from a social science
background so it's kind of like for me
it was just kind of this plunge into

talking about like U statistical physics
and so on that it's like it's super
interesting once it it starts to catch
on but until that point it's I'm just
like thinking like how do I use this
right so that's that's exactly what I'm
trying to get at but yeah no stream
sounds really awesome so I'd be happy to
do that cool okay thank you Andrew and
cool to hear the update thank you all
for joining see you next time take care